PROJECT PROPOSAL

SkillNet: A Service-Oriented Multi-Tier Job Hiring & Service Portal

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1. Project Title

SkillNet: A Service-Oriented Multi-Tier Job Hiring & Service Portal

2. Introduction

The rapid growth of the gig economy and digital recruitment platforms has transformed how organizations and individuals connect with skilled workers. However, existing job portals primarily focus on either corporate employment opportunities or informal labor services, creating a gap between employers, service providers, and customers. Additionally, many independent workers are required to publicly share their personal contact information to get jobs, exposing them to privacy and security risks.

To address these challenges, this project proposes **SkillNet**, a Service-Oriented Programming (SOP)-based platform that integrates corporate recruitment and local service hiring into a single secure ecosystem. The system enables businesses to identify qualified candidates efficiently through an automated matching engine while allowing local customers to locate nearby independent skilled workers (such as painters, masons, and electricians) as guest users. SkillNet prioritizes user privacy and security by abstracting direct contact details and routing initial service inquiries through an automated, secure notification ecosystem.

3. Background

Finding suitable talent remains a significant challenge for both corporate organizations and individuals seeking daily-wage services. Corporate HR departments spend considerable time reviewing resumes manually, while local customers often struggle to find reliable, skilled workers within their exact geographic locality.

At the same time, independent workers frequently advertise their services through social media or public notice boards, exposing their personal phone numbers and email addresses. This creates opportunities for spam, fraud, and misuse of personal information. SkillNet aims to solve these issues by introducing an integrated platform that combines intelligent recruitment, localized service discovery, and privacy-preserving automated routing using a modular Service-Oriented Architecture.

4. Problem Statement

The current recruitment and service-hiring ecosystem faces several distinct challenges:

- **4.1 Hiring Difficulties:** Local customers experience delays and friction when searching for nearby skilled independent workers due to the lack of a centralized, location-aware service directory.
- **4.2 Privacy and Security Concerns:** Independent workers are compelled to share personal contact credentials publicly to gain visibility, increasing the risk of spam, scams, and unauthorized data harvesting.
- **4.3 Inefficient Corporate Recruitment:** Corporate HR teams spend significant effort manually filtering candidates based on experience, specific skill alignments, and location requirements.

5. Objectives

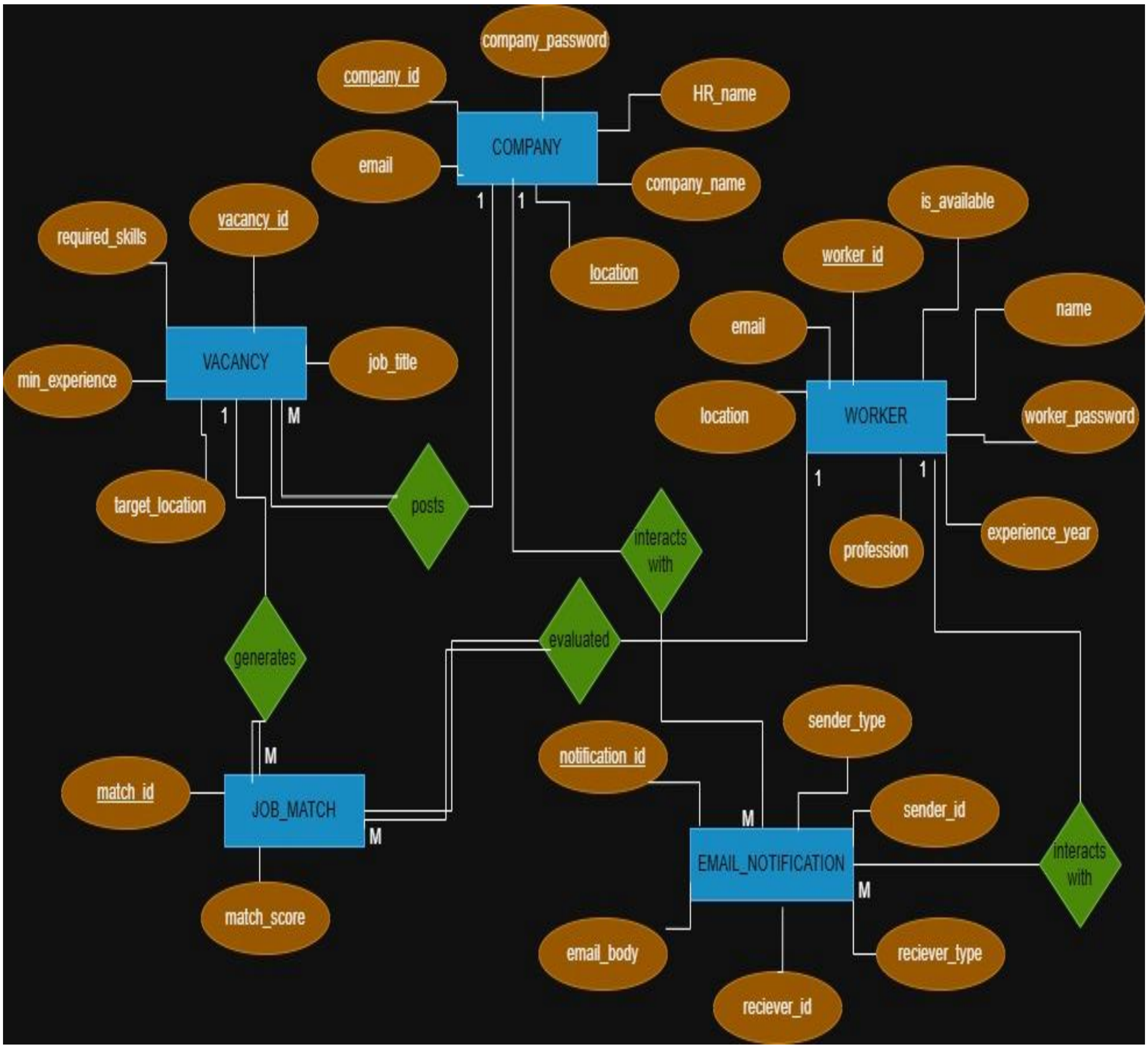
The primary objectives of SkillNet are:

- To design and implement a modular web system using Service-Oriented Programming (SOP) principles.
- To implement a secure, role-based authentication mechanism to protect corporate dashboards and worker configurations.
- To develop an automated HR Filtering and Matching Engine that calculates match scores for corporate vacancies based on predefined criteria.
- To protect worker privacy by eliminating public contact listings and routing initial guest customer requests through a secure, automated backend notification service.

- To provide a real-time Availability Toggle for independent workers to indicate immediate job readiness.

6.DataBase Implementation

EER Diagram



Schema diagram

7. Proposed System & Microservices Architecture

Following the SOP paradigm, the system is decoupled into three independent, full-stack microservices communicating seamlessly through secure RESTful APIs:

Microservice 1: User, Worker Profile & Security Service

- **Role-Based Authentication:** Implements secure registration and login mechanisms for Corporate HR and Independent Workers using encrypted credentials (company_password and worker_password) to isolate dashboard roles.
- **Worker Profile Management:** Enables independent service providers to build professional profiles containing their name, verified email, profession (e.g., painter, electrician), years of experience, and location.
- **Availability Toggle:** A real-time backend toggle allowing self-employed workers to update their status as Available or Unavailable, dynamically managing their visibility to local search queries.

Microservice 2: Corporate Job & Vacancy Service

- **Vacancy Management:** A dedicated interface for authenticated corporate HR personnel to create job vacancies, define precise recruitment criteria, and specify requirements for target job titles, required skills, minimum experience years, and locations.

Microservice 3: Intelligent Matching & Secure Notification Service

- **Automated Matching Engine:** A specialized backend algorithm that automatically queries candidate profiles against posted corporate vacancy requirements, executing dynamic score calculations to generate a ranked candidate recommendation list (match_score).

- **Localized Guest Discovery:** Allows non-registered guest customers to instantly scan for local nearby workers based on profession and area criteria.
- **Privacy Protection via Automated Notifications:** Obscures direct personal worker contact details from public views. Initial customer inquiries and corporate job shortlists instantly trigger the automated system backend to generate secure, transactional logs and dispatch directed alerts to the worker's private inbox (EMAIL_NOTIFICATION) without raw database correlation exposure.

8. System Architecture

The application follows a Service-Oriented Architecture (SOA) where each full-stack layer operates independently:

- **Presentation Layer:** React.js frontend providing responsive, clean user interfaces and modular dashboards tailored for each user role.
- **Application Layer:** Spring Boot backend services containing core business logic, matching algorithms, cryptographic validation, and REST APIs.
- **Data Layer:** SQL relational database managed cleanly without redundant timestamps or complex security hashes for efficient initial prototyping.

9. Technology Stack

Component	Technology
Frontend	React.js
Backend	Spring Boot
Database	MySQL / SQL
API Testing	Postman
Architecture	Service-Oriented Programming (SOP) / Microservices

10. Expected Outcomes

Upon successful completion, SkillNet will provide:

- A fully functional web application demonstrating true service-oriented modularity across decoupled services.
- An intelligent matching mechanism reducing time-to-hire and automated filtration load for corporate recruitment teams.
- A frictionless, localized service discovery tool granting guest customers instant access to independent workers.
- Enhanced data privacy for independent manual laborers by keeping initial contact stages entirely automated and abstracted through secure communication triggers.

11. Conclusion

SkillNet delivers a secure, modern answer to contemporary recruitment and local service-hiring vulnerabilities. By synthesizing Service-Oriented Architecture, structured data filtration, and strict privacy abstraction, the platform optimizes workforce engagement for corporate entities, customers, and independent service providers alike.